

SYNENT Technology Internship

Task 2: Number Guessing Game

1. Introduction

The Number Guessing Game is a beginner-friendly command-line application developed in Python as part of the SYNENT Technology Internship. The program generates a random number between 1 and 100 and asks the player to guess it.

2. Objective

The objective is to demonstrate random number generation, user input, loops, conditional statements, comparison operators, and basic exception handling.

3. Technologies Used

Technology	Purpose
Python 3	Programming language used to develop the game
random module	Generates the secret number
CLI / Terminal	Provides user interaction and output

4. Methodology

- The program imports Python's random module.
- A random secret number from 1 to 100 is generated.
- The player enters guesses inside a while loop.
- The program compares each guess with the secret number.
- Higher and lower hints are displayed to guide the player.
- The loop ends when the correct number is guessed.
- Invalid non-numeric input is handled with an error message.

5. Features

• Random number generation • Repeated guesses • Higher/lower hints • Success message • Invalid-input handling

6. Sample Output

```
===== NUMBER GUESSING GAME =====
Guess a number between 1 and 100.
Enter your guess: 50
Too high! Try a smaller number.
Enter your guess: 25
Too low! Try a larger number.
Enter your guess: 37
Congratulations! You guessed the correct number.
```

7. Result

The game successfully generates a random number and provides useful hints until the player guesses the correct number.

8. Conclusion

This project demonstrates practical use of Python fundamentals, especially random number generation, loops, conditional logic, input handling, and basic error handling.